Fiscal Year:	FY 2018	**Task Last Updated:**	FY 12/20/2017
PI Name:	Hyers, Robert Ph.D.		
Project Title:	Unified Support for THERMOLAB - ISS, ICOPROSOL, and PARSEC (NNX16AB40G)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Worcester	**State:**	MA
Zip Code:	01609-2280	**Congressional District:**	2
Comments:	Formerly with University of Massachusetts at Amherst.		
Project Type:	Flight	**Solicitation / Funding Source:**	2009 ESA AO Biology on Sounding Rockets and Physical Sciences on All Platforms
Start Date:	11/16/2015	**End Date:**	11/15/2020
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA MSFC
Contact Monitor:	Craven, Paul	**Contact Phone:**	256-544-7649
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:	Dr. Jonghyun Lee joined as Co-Investigator in 2016 to assist with the growing needs for computational support for different experiments reflecting the broadening acceptance of the value added by our calculations. He left UMass to join the faculty of Iowa State University as a tenure-track Assistant Professor in September 2017.		
COI Name (Institution):	Lee, Jonghyun Ph.D. (UMass Amherst)		
Grant/Contract No.:	NNX16AB40G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

NOTE: Continuation of "Unified Support for THERMOLAB - ISS, ICOPROSOL, and PARSEC," grant NNX10AR71G with the same investigator, Dr. Robert Hyers.
Robert Hyers is U.S. Team Member on European Space Agencey (ESA)-sponsored projects, "Peritectic Alloy Rapid Solidification with Electromagnetic Convection (PARSEC)," "Thermophysical Properties of Liquid Metallic Alloys - Modelling of Industrial Solidification Processes and Development of Advanced Products (THERMOLAB - ISS)," and "Thermophysical properties and solidification behaviour of undercooled Ti-Zr-Ni liquids showing an icosahedral short-range order (ICOPROSOL)."

Funding is for Dr. Hyers's role as Team Member on the subprojects described below, associated with European Space Agency (ESA)-sponsored main projects as noted.

Main project: Peritectic Alloy Rapid Solidification with Electromagnetic Convection (PARSEC); Science Coordinator is Dr. Thomas Volkmann, German Aerospace Center DLR, Institut für Materialphysik im Weltraum, Linder Höhe, Köln, Germany

Subprojects:

1. Transformation Kinetics-ISS (International Space Station). Task description: Using results from thermophysical property evaluation and combining these with the Magnetohydrodynamic (MHD) convection definition, we will set convection over as wide a range as is possible and conduct recalescence studies from the undercooled melt using the Material Science Laboratory Electromagnetic Levitator (MSL-EML) to compare to results from ground-based EML and ESL. The critical parameters of interest are evaluation of the incubation time between formation of metastable phases and subsequent transformation to the stable phase for Fe-based and titanium aluminide alloys. Since this transformation is very rapid, the high-speed video system must be used to image the processes associated with nucleation and growth during phase selection.

2. Thermophysical Properties. Task description: The density, viscosity, and surface tension of the flight samples will be measured in flight for a range of temperatures. Density will be determined by the video metrology, and surface tension and viscosity by the oscillating drop method. This WP includes determination of parameter modifications required for these measurements as well as collection, reduction, and analysis of the resulting data. The resulting data will support the whole range of theoretical and modeling efforts in this proposal.

3. Modeling of Macroconvection. Task description: Magnetohydrodynamic (MHD) model of macroconvection in the samples for different combinations of experimental parameters. Evaluation of fluid velocity, shear rates, turbulence intensity, and their distribution for control of experiments and input to other theoretical and modeling efforts. Collaboration on development of parameters for other work packages to enable the use of convective state as an experimental parameter.

Main project: Thermophysical properties and solidification behaviour of undercooled Ti-Zr-Ni liquids showing an icosahedral short-range order (ICOPROSOL); Science Coordinator is Dr. Dirk Holland-Moritz, Deutsches Zentrum für Luft- und Raumfahrt (DLR), Institut für Materialphysik im Weltraum, Linder Höhe, Köln, Germany.

Subproject:

Modeling of macroconvection: Magnetohydrodynamic modelling of macroconvection in Ti-Zr-Ni samples for different combinations of experimental parameters. Evaluation of fluid velocity, shear rates, turbulence intensity, and their distribution for control of experiments and input to other theoretical and modeling efforts.

Main project: Thermophysical Properties of Liquid Metallic Alloys - Modelling of Industrial Solidification Processes and Development of Advanced Products (THERMOLAB - ISS); Science Coordinator is Professor Hans-Joerg Fecht, Ulm University, Ulm, Germany

Subprojects:

Long Duration Microgravity Experiments - ISS. Evaluation of microgravity experiments for fluid flow investigations.

MODELLING A: Thermodynamics, Phase Selection. Modelling of nucleation under various flow conditions for steel alloys.

MODELLING B: Non-contact thermophysical. Modeling of magnetohydrodynamic convection and shear in 1-g, microgravity for 4-6 alloys. Modeling of Marangoni convection in 1-g ESL drops for comparison with microgravity experiments. Modeling of temperature distribution in 1-g and microgravity samples, 4-6 alloys. Input into design of experimental parameters for 4-6 Alloys.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

This project consists of modeling in support of three different flight experiments:
- PARSEC - Peritectic Alloy Rapid Solidification with EM Convection

- ThermoLAB - Thermophysical Properties of Liquid Metallic Alloys

- ICOPROSOL - Thermophysical properties and solidification behaviour of undercooled Ti-Zr-Ni liquids showing an icosahedral short-range order

In PARSEC, we are investigating the effect of fluid flow on the solidification path of certain alloys. Control of the solidification path would enable tailoring of the microstructure and properties of metal parts for specific applications.

In ThermoLAB, we are investigating the thermophysical properties of high-temperature materials. Many of these materials are used commercially. A better understanding of the physical properties will allow more efficient and more reliable production of metallic parts using these alloys. Other alloys are chosen because of scientific interest in their properties, which may illuminate fundamental physics related to the formation of bulk metallic glasses and quasicrystals.

In ICOPROSOL, we are investigating the nucleation and growth of quasicrystals, and the effect of atomic-scale order on the macroscopic properties of these alloys. This fundamental investigation may improve our ability to tailor the

	microstructure of metals for commercial applications.
Task Progress:	• Supported the magnetohydrodynamic calculations for the prediction of phase selection during solidification for PARSEC. • Began the magnetohydrodynamic calculations for analysis of the transition to turbulence in positioner-dominated flow in MSL-EML for SEMITHERM. • Began revised MHD calculations for QUASI and ICOPROSOL. • Visited DLR Cologne August 30, 2017, for planning of Batch 3 parameters and analysis of anomalous nucleation events observed in Batch 1. This visit led to the discovery of more anomalous nucleation events. Analysis is ongoing. • Measured thermophysical properties of Zr-O using ESL at NASA MSFC (Marshall Space Flight Center) in support of ISS experiments. • Co-edited special journal issue of JOM on Materials Research in Microgravity. • J. Lee edited the proceedings volume Materials Processing Fundamentals 2017. • Hired Ms. Gwendolyn Bracker, B.S. Materials Science and Engineering from Case Western. • Attended International Work Group Meeting on MSL-EML at ESA ESTEC, Noordwijk, Netherlands, August 28-29, 2017. • Image processing code for the analysis of MSL-EML data is being adjusted (expect to process the data downloaded from ISS this fall).
Bibliography Type:	Description: (Last Updated: 03/30/2026)
Articles in Peer-reviewed Journals	Matson DM, Xiao X, Rodriguez JE, Lee J, Hyers RH, Shuleshova O, Kaban I, Schneider S, Karrasch C, Burggraff S, Wunderlich R, Fecht H-J. "Use of thermophysical properties to select and control convection during rapid solidification of steel alloys using electromagnetic levitation on the Space Station." JOM (Journal of The Minerals, Metals & Materials Society). 2017 Aug;68(8):1311-8. https://doi.org/10.1007/s11837-017-2396-5 , Aug-2017
Articles in Peer-reviewed Journals	Lee J, SanSoucie MP. "Experiments using a ground-based electrostatic levitator and numerical modeling of melt convection for the iron-cobalt system in support of space experiments." JJOM (Journal of The Minerals, Metals & Materials Society). 2017 Aug;69(8):1298-302. https://doi.org/10.1007/s11837-017-2387-6 , Aug-2017
Articles in Peer-reviewed Journals	Matson DM, Hyers RW, Genau A, Zaeem MA. "Solidification behavior in reduced gravity." JOM (Journal of The Minerals, Metals & Materials Society). 2017 Aug;69(8):1258-60. https://doi.org/10.1007/s11837-017-2418-3 , Aug-2017
Books/Book Chapters	Zhao J, Lee J, Wunderlich R, Fecht H-J, Schneider S, SanSoucie M, Rogers JR, Hyers RW. "Influence of Oxygen on Surface Tension of Zirconium." in "Materials Processing Fundamentals 2017. The Minerals, Metals & Materials Series." Ed. A. Allanore, G. Lambotte, J. Lee. Cham: Springer International Publishing, 2017. p, 47-54. https://doi.org/10.1007/978-3-319-51580-9_5 , Mar-2017
Books/Book Chapters	Allanore A. Lambotte G, Lee J. (Eds.) "Materials Processing Fundamentals 2017." Ed. A. Allanore, G. Lambotte, J. Lee. Cham: Springer International Publishing, 2017. https://doi.org/10.1007/978-3-319-51580-9 , Mar-2017